

ALBUM SALES & FAN ENGAGEMENT DASHBOARD

Project Report

Tools Utilized: Microsoft Excel, Power Query
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TABLE OF CONTENTS

S.No.	Title	Page No
1.	Executive Summary	3
2.	Project Objectives & Scope	4
3.	Data Engineering & Methodology	5-6
	3.1 ETL Process using Power Query	
	3.2 M-Code Implementation & Advanced Logic	
	3.3 Data Normalization	
	3.4 Analytical Workflow Summary	
	3.5 Dashboard Integration & Automation	
4.	Workbook Structure	6
5.	The Fan Engagement Score (FES)	7-8
	5.1 Components of FES	
	5.2 Metric Normalization	
	5.3 The Weight Control Panel	
	5.4 FES Calculation Model	
	5.5 Tier Classification System	
	5.6 Key FES Insights	
6.	Market Segment Analysis(Global vs Korea)	9
	6.1 The Export Index	
	6.2 Market Classification	
	6.3 Geographic Insights	
7.	Streaming & Digital Consumption Intelligence	10
	7.1 Platform Dominance & Market Share	
	7.2 The Stream-to-Sale Ratio	
	7.3 Audience Behavior & Platform Insights	
	7.4 Strategic Insights	
8.	Artist Lifecycle & Momentum Modeling	11
	8.1 Historical Growth Index(Momentum)	
	8.2 Lifecycle Distribution	
	8.3 Peak Performance Timing	
	8.4 Strategic Insights	
9.	Label Performance & Market Share	12
	9.1 Label Portfolio Distribution	
	9.2 Key Performance Insights by Label	
	9.3 Strategic Market Concentration	
10.	Dashboard Architecture & Technical Implementation	13-14
	10.1 Dashboard Structure	
	10.2 Key Visualization Elements	
	10.3 Interactive Features	
	10.4 Automation & Scalability	
11.	Conclusion & Future Outlook	15
	11.1 Conclusion	
	11.2 Key Findings	
	11.3 Future Outlook	

1.Executive Summary

This project presents a comprehensive data analytics solution designed to evaluate **Album Sales & Fan Engagement** trends within the music industry from 2019 to 2026. The objective of the project is to transform fragmented data from multiple platforms into a unified, interactive dashboard that enables meaningful insights and data-driven decision-making.

The dataset captures large-scale industry activity, tracking over **800 albums across 190 unique artists**. Over the observed period, the industry generated approximately **408.59 million physical album sales** and **2.81 trillion digital streams** across major platforms.

A key finding of this project is the significant shift toward **globalized consumption**. With an average **Export Index of 75.3%**, international markets have become the primary driver of growth rather than a secondary segment. Leading artists such as Seventeen (top sales performance) and BTS (highest engagement levels) demonstrate the scale of global influence, while platform analysis reveals that **Spotify dominates the streaming ecosystem**, accounting for nearly 80% of total digital consumption.

To convert raw data into actionable intelligence, the project introduces a set of advanced analytical frameworks. The **Fan Engagement Score (FES)** is a core innovation, representing a weighted and normalized metric (0–100) that integrates streaming activity, social engagement, and physical sales into a single performance indicator. This allows for consistent comparison and ranking across artists and albums. Additionally, a **Loyalty Classification model** evaluates fan behaviour through the stream-to-sale ratio, distinguishing between *stream-heavy* and *physically loyal* audiences, while a **Historical Growth Index(Momentum)** categorizes artist lifecycle stages based on historical performance trends.

Key insights from the analysis indicate that while streaming platforms significantly expand audience reach, they do not always translate proportionally into physical sales, highlighting differences in fan behaviour and monetization patterns. Furthermore, the industry exhibits clear segmentation between globally focused and domestically dominant artists and labels, reflecting diverse strategic approaches to market positioning.

Overall, this project demonstrates how structured data engineering, advanced metric design, and interactive visualization can be combined to deliver a scalable and executive-ready analytics solution. The resulting dashboard serves as a powerful decision-support tool for stakeholders in marketing, artist management, and global distribution, with potential for future expansion into predictive analytics and real-time data integration.

2. Project Objectives & Scope

2.1 Project Objectives

The primary objective of this project is to design and develop a comprehensive, interactive analytics dashboard that evaluates album sales performance and fan engagement across multiple platforms within the music industry.

The specific objectives of the project are as follows:

- To analyse album sales trends from 2019 to 2026 and identify peak performance periods
- To integrate actual projections for 2024–2026, providing a forward-looking perspective on industry trends rather than just a historical look-back.
- To evaluate fan engagement across digital platforms including streaming services and social media channels
- To develop a unified performance metric, namely the **Fan Engagement Score (FES)**, to enable consistent comparison across artists and albums
- To assess the relationship between streaming activity and physical album sales
- To analyse global versus domestic (Korea) market performance and identify export-driven growth patterns
- To evaluate label-level performance and understand strategic positioning within the industry
- To build an interactive Excel-based dashboard that supports dynamic filtering, comparison, and visualization of key metrics

2.2 Project Scope

The scope of this project encompasses a multi-dimensional analysis of music industry performance across a defined timeframe, dataset scale, and platform coverage.

Scope of Analysis

- **Timeframe:** The study spans a multi-year period from **2019 to 2026**, enabling the analysis of long-term trends and performance patterns.
- **Dataset Volume:** The analysis includes **800 individual albums across 190 unique artists**, providing a comprehensive representation of industry output.
- **Market Coverage:** The project evaluates performance across **37 record labels**, including both major industry leaders and emerging independent firms, allowing for comparative analysis of market dominance and strategic positioning.
- **Data Channels:** The study integrates cross-platform metrics, including:
 - Physical album sales
 - Spotify Global streaming data
 - YouTube viewership
 - Domestic platform engagement via Melon Korea

3. Data Engineering & Methodology

The technical foundation of this project is built upon a structured data analytics pipeline leveraging a high-performance **ETL (Extract, Transform, Load) architecture**. The methodology focuses on ensuring **data integrity, structural simplification, and efficient processing** of large-scale music industry datasets.

The analytical workflow is designed to transform fragmented multi-source data into a unified, scalable model capable of supporting dynamic dashboard interactions and advanced metric calculations.

3.1 ETL Process using Power Query

The project utilizes **Excel Power Query (M-Code)** to implement a robust ETL pipeline.

Extract

Raw data, including historical records and projected values (2019–2026), was imported into the Power Query environment.

Transform

Advanced transformation logic was applied, including:

- Removal of null and duplicate values
- Standardization of formats (dates, numeric values)
- Data type corrections for accurate computation
- Creation of conditional columns for:
 - Artist Momentum classification
 - Loyalty Class segmentation

3.2 M-Code Implementation & Advanced Logic

To enhance analytical precision, custom **M-Code transformations** were implemented:

- **Artist-Level Filtering:**
Logic was developed to identify key attributes such as the top-performing album for each artist by selecting rows corresponding to maximum sales values within grouped datasets.
- **Data Normalization:**
Streaming metrics from Spotify, YouTube, and Melon, along with physical sales data, were transformed into a common scale to enable fair comparison and integration into the Fan Engagement Score.
- **Metric Synthesis:**
Data from **800 albums and 190 artists** was merged into a unified structure, enabling cross-dimensional analysis of sales, streaming, and engagement.

3.3 Data Normalization

Due to differences in scale across metrics (e.g., streaming counts vs social engagement), normalization techniques were applied. This step is essential for constructing the Fan Engagement Score (FES).

Each metric was scaled to a **0–1 range**, ensuring:

- Consistency across variables and Accurate weighting in composite calculations

3.4 Analytical Workflow Summary

The overall workflow follows:

- **Extraction:** Importing raw datasets into Power Query
- **Transformation:** Applying cleaning, normalization, and classification logic
- **Loading:** Connecting the Master Table to the data model
- **Analysis:** Using Pivot Tables and calculated metrics
- **Visualization:** Building interactive dashboards

3.5 Dashboard Integration & Automation

The final processed dataset is integrated into interactive dashboards using:

- Pivot Tables for aggregation
- Slicers for dynamic filtering
- Conditional formatting for performance visualization
- KPI cards for summary metrics

All dashboard components are dynamically linked to the Master Table, ensuring **automatic updates and real-time responsiveness** when data changes.

4. Workbook Structure

Sheet Name	Layer	Description
Master Data	Data Engine	The primary "Single Source of Truth" resulting from the ETL process; contains all 800 albums and 190 artists with standardized formats.
Artist level	Data Engine	A specialized view of the Master Table grouped by Artist ID, used to feed artist-specific performance slicers and ranking charts.
Label wise	Data Engine	A subset of the Master Table organized by Record Label (37 unique entities) to facilitate portfolio-level market share analysis.
Dashboard	Visualization	The central executive interface featuring high-level KPI cards, the Artist Tier Distribution, and global interactive slicers.
Fan Engagement	Visualization	A focused report on the Fan Engagement Score metrics, illustrating the balance between reach, engagement, and loyalty.
Streaming	Visualization	A dedicated visualization interface focusing on platform market share (Spotify, YouTube, Melon) and digital consumption trends.
Global vs Korea	Visualization	A specialized analytics view comparing domestic South Korean performance against international export metrics and the Export Index.
Label Analysis	Visualization	A comparative dashboard focused on label efficiency, artist distribution, and total volume by record label.
Cals	Processing	The backend "Engine Room" containing Pivot Tables, Charts and reference calculations that drive all dynamic dashboard visualizations.

5. The Fan Engagement Score (FES)

To effectively evaluate artist performance across multiple dimensions, this project introduces a composite metric known as the **Fan Engagement Score (FES)**.

Traditional metrics such as album sales or streaming counts independently fail to capture the complete picture of artist success. Therefore, the FES model is designed to integrate **sales performance, digital streaming activity, and fan engagement metrics** into a single standardized score.

The FES provides a **holistic measure of artist and album performance**, enabling consistent comparison across different scales and platforms.

5.1 Components of FES

The Fan Engagement Score is constructed using three primary data dimensions:

- **Physical Performance**
 - Total album sales (Global + Korea)
- **Digital Engagement**
 - Spotify streams
 - YouTube views
 - Melon streams
- **Fan Interaction Metrics**
 - Twitter mentions
 - Weverse posts
 - Fan Cafe membership

5.2 Metric Normalization

Since the selected metrics vary significantly in scale, normalization is applied to ensure comparability.

Each metric is transformed into a standardized range between **0 and 1**, using relative maximum-based scaling:

$$\text{Normalized Value} = \text{Metric Value} / \text{Maximum Value}$$

This ensures that no single metric disproportionately influences the final score due to scale differences.

5.3 The Weight Control Panel

To reflect the relative importance of each component, weights are assigned to different categories of metrics.

The weighting structure is designed as follows:

- **Streaming Metrics:** High weight due to large-scale audience reach
- **Fan Engagement Metrics:** Moderate weight indicating community strength
- **Sales Metrics:** Moderate weight representing direct revenue

WEIGHT CONTROL PANEL			
Code	Metric	Weight ⬆️	What it measures
W1	Spotify Streams	0.18	Global audio streaming — international reach
W2	YouTube MV Views	0.17	Video engagement — visual content consumption
W3	Melon Streams	0.15	Korea domestic streaming — local fan loyalty
W4	Twitter Mentions	0.14	Social buzz — real-time cultural conversation
W5	Weverse Posts	0.13	Official fan platform — dedicated fandom activity
W6	Fan Cafe Members	0.12	Registered community — long-term fan depth
W7	Album Sales	0.11	Physical purchase — strongest loyalty signal
WEIGHT TOTAL		1.00	☒ Valid — FES calibrated

5.4 FES Calculation Model

To ensure mathematical consistency across variables with vastly different scales—such as billions of streams versus thousands of community members—a **Min-Max Scaling** normalization technique was applied before the final weighted summation.

The FES Formula:

The Fan Engagement Score is computed as a weighted sum of normalized metrics:

$$FES = \sum_{i=0}^n (W_i * \text{Normalized Value})$$

The final score is scaled to a **0–100 range** for ease of interpretation.

$$\text{Normalized FES} = FES * 100$$

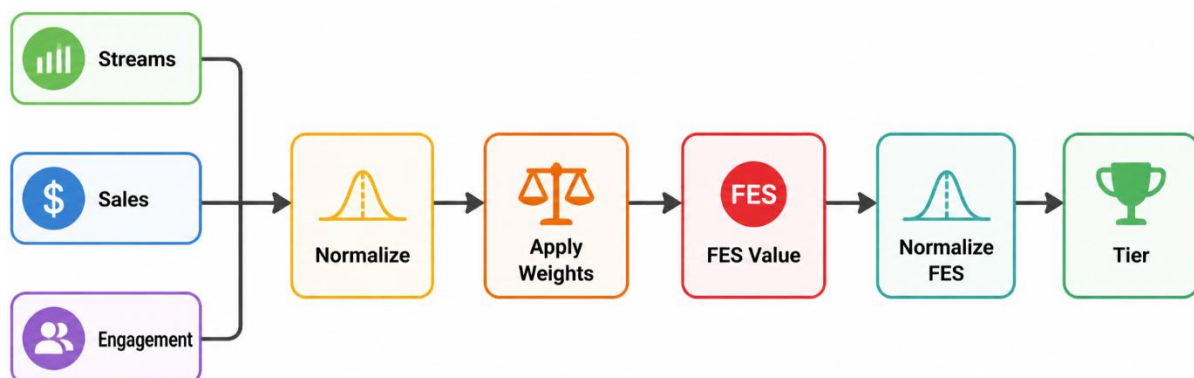
5.5 Tier Classification System

To simplify analysis, albums are automatically categorized into a five-tier hierarchy based on their **Normalized FES**. This classification enables quick segmentation and comparative analysis.

- **LEGEND (>=75):** Exceptional performance across all metrics
- **ICONIC (>=60):** Strong performance with high engagement
- **RISING (>=40):** Moderate growth and visibility
- **SOLID (>=20):** Stable but limited impact
- **EMERGING (<20):** Low engagement and performance

5.6 Key FES Insights

- **Benchmark Leader:** BTS remains the industry gold standard, holding the record for the highest overall FES.
- **Top Performing Album:** *LOVE YOURSELF 轉 'Tear'* achieved the highest FES score in the dataset.
- **Tier Concentration:** While the Solid tier contains the largest volume of artists (444), only 12 artists currently hold the Legend status, highlighting the extreme concentration of high-tier fan engagement.



6. MARKET SEGMENT ANALYSIS (GLOBAL vs KOREA)

This section analyses the distribution of album sales across **domestic (Korea)** and **international (Global)** markets, providing insights into the evolving geographic dynamics of the music industry.

With the rapid expansion of global streaming platforms and international fanbases, understanding the balance between domestic and global consumption is critical for evaluating artist performance and market strategy.

6.1 The Export Index

A critical KPI developed for this project is the **Export Index**, which measures the percentage of an artist's total sales and engagement generated outside of South Korea.

- **Industry Average:** The dataset reveals an average Export Index of **75.3%**, signalling that international markets now contribute three-quarters of the industry's primary momentum.
- **Market Penetration:** High Export Indexes are strongly correlated with artists categorized in the "Legend" and "Iconic" FES tiers.

6.2 Market Classification

Based on sales distribution and Export Index values, artists and albums are categorized into three market types:

- **Balanced (49%):** The largest segment, representing artists with stable, synchronized support in both domestic and international territories.
- **Global-Led (24%):** Artists whose primary growth is driven by international fans, often showing higher Spotify-to-Melon streaming ratios.
- **Korea-Led (27%):** Artists who maintain a strong domestic stronghold but have yet to achieve significant global breakout volume.

6.3 Geographic Insights

The analysis distinguishes between domestic "local" success and international "export" success:

- **Top Artist by Global Sales:** Seventeen.
- **Top 10 Global Artists:** Includes high-export performers like **ATEEZ** and **aespa**, who show disproportionately high engagement scores in international territories compared to their domestic baseline.
- **Sales Volume:** Total historical global sales across the dataset reached **198,251,330 units**, nearly reaching parity with the **210,339,271 units** sold within Korea.

7. Streaming & Digital Consumption Intelligence

This section examines the role of digital streaming platforms in shaping music consumption patterns and artist performance. With the rise of global streaming services, digital engagement has become a primary driver of audience reach and visibility.

The analysis focuses on understanding **platform dominance, audience behaviour, and the relationship between streaming activity and physical album sales.**

7.1 Platform Dominance & Market Share

The dataset tracks a total of **2.81 trillion digital streams**, revealing a heavily centralized digital ecosystem.

- **Spotify Global:** Acts as the primary driver of international reach, commanding an **80% platform share** of the tracked streaming market.
- **YouTube:** Remains the critical medium for visual consumption, totalling **1.59 trillion views** and serving as a gateway for viral artist discovery.
- **Melon Korea:** Representing **19%** of the tracked audio streaming market, it remains the gold standard for measuring domestic South Korean popularity and General Public appeal.

7.2 The Stream-to-Sale Ratio

To understand fan loyalty, a custom **Stream : Sale Ratio** was developed.

$$\text{Stream-to-Sale Ratio} = \text{Total Streams} / \text{Total Sales}$$

This metric provides insights into fan behaviour and purchasing patterns.

- **Physically Loyal:** Artists with a low ratio (high sales, lower streams). These fanbases exhibit high financial commitment but may have lower daily listening ubiquity.
- **Stream-Heavy:** Artists with a high ratio. These acts often have viral digital hits but lower conversion rates into high-value physical purchases.

7.3 Audience Behaviour & Platform Insights

The analysis reveals distinct patterns in audience behaviour:

- Global audiences primarily engage through streaming platforms
- Domestic audiences show stronger loyalty through both streaming and physical purchases
- Platform preference varies based on geography and artist reach

These patterns demonstrate that **digital consumption behaviour is not uniform**, but influenced by regional and platform-specific dynamics.

7.4 Strategic Insights

The streaming analysis provides several important strategic conclusions:

- Streaming platforms are essential for **global reach and discovery**
- High streaming numbers do not always translate into direct revenue through album sales
- A balanced strategy combining digital engagement and physical sales is crucial for long-term success
- Platform-specific strategies can enhance audience targeting and engagement

8. Artist Lifecycle & Momentum Modeling

This section focuses on analyzing the **evolution of artist performance over time**, using historical data to identify growth patterns, stability, and decline phases within an artist's career lifecycle.

Unlike static performance metrics, lifecycle modeling provides a **temporal perspective**, enabling a deeper understanding of how artists develop, peak, and sustain their influence in the industry.

8.1 Historical Growth Index (Momentum)

To determine if an artist is gaining or losing market share, a custom **Growth Index** was programmed into the Master Table. This index measures the delta between an artist's debut project metrics and their current era performance.

Artists are assigned one of three **Momentum Classifications**:

- **Accelerating:** Indicates exponential growth, where current project performance significantly exceeds historical baselines.
- **Growing:** Represents steady, sustainable upward trends across both physical and digital platforms.
- **Softening:** Identifies artists whose growth has stabilized or entered a natural lifecycle plateau. *Note: For "Legend" tier artists, a softening tag often indicates market saturation rather than a decline in relevance.*

8.2 Lifecycle Distribution

The dashboard utilizes the **Artist Tier** system to map the industry lifecycle. Analysis of the 190 unique artists reveals a pyramid structure:

- **The Elite Core:** Only **12 artists** have achieved the "Legend" classification, representing the peak of the industry lifecycle.
- **The Growth Sector:** The **Rising** tier consists of **115 artists**, serving as the primary engine for future industry expansion.
- **The Industry Foundation:** The **Solid** (444 artists) and **Emerging** (198 artists) categories provide the necessary volume for label portfolio diversification.

8.3 Peak Performance Timing

The modeling identifies critical windows for artist success:

- **Peak Industry Year: 2023** stands as the definitive high-water mark for artist lifecycle performance across the dataset.
- **Optimal Release Window: May** is identified as the month where artists typically see the highest engagement and sales growth index.

8.4 Strategic Insights

The lifecycle and momentum analysis reveals several key insights:

- Artist success is dynamic and evolves over time rather than remaining constant
- High-performing artists may still experience declining momentum
- Emerging artists with accelerating growth represent significant future opportunities
- Combining historical trends with current performance provides a more accurate evaluation of artist potential

9. Label Performance & Market Share

This section analyzes the role of record labels in shaping industry performance, focusing on **market share, artist portfolios, and engagement efficiency**. Labels act as key strategic entities responsible for artist development, distribution, and market expansion.

Understanding label-level performance provides insights into **industry dominance, competitive positioning, and strategic direction**.

9.1 Label Portfolio Distribution

The project tracks 190 unique artists distributed across 37 labels, revealing a diverse range of business models.

Each label manages a portfolio of artists with varying levels of performance and engagement.

Key observations include:

- Labels with a **diversified artist portfolio** tend to maintain stable performance across market fluctuations
- Labels heavily dependent on a few top artists exhibit higher risk but potentially higher peak performance
- Strong portfolios are characterized by a mix of:
 - High-performing (Legend/Iconic) artists
 - Emerging (Rising) talent

This analysis demonstrates how portfolio composition impacts **long-term sustainability and growth**.

9.2 Key Performance Insights by Label

By aggregating artist FES and sales data at the label level, the following market leaders were identified:

- **Highest Total Sales by Label:** HYBE leads the industry in total physical sales volume, driven by high-output artists like Seventeen and BTS.
- **Highest Digital Engagement Portfolio:** YG Entertainment and SM Entertainment show high concentrations of digital streaming and YouTube engagement relative to their artist count.
- **Portfolio Efficiency:** This metric measures the average FES per artist. Labels with fewer, higher-performing acts (e.g., JYP Entertainment) show high efficiency compared to labels with larger, less-active artist rosters.

9.3 Strategic Market Concentration

The dashboard's "Label Portfolio" analysis demonstrates a tiered market structure:

- **Top 5 Labels:** Command over **60% of the total digital streaming market share**.
- **Market Fragmentation:** Despite the dominance of major players, the 2024–2026 projections suggest a growing market share for mid-tier labels as global sub-genres continue to diversify.

10. Dashboard Architecture & Technical Implementation

The final phase of this project involved translating the engineered Master Table into a high-performance, interactive visual interface. The dashboard is designed as an interactive analytical tool that provides a comprehensive view of album sales, fan engagement, streaming performance, and market dynamics. It is structured into multiple pages, each focusing on a specific analytical perspective. The design emphasizes:

- Clarity and readability
- Interactive filtering
- Logical data flow
- Insight-driven visualization

10.1 Dashboard Structure

The dashboard consists of five main analytical modules:

- **Overview Dashboard:** Provides a high-level summary of industry performance, including total sales, total streams, top artist, and trend analysis
- **Fan Engagement Dashboard:** Focuses on the Fan Engagement Score (FES), tier distribution, and top-performing artists
- **Streaming Dashboard:** Analyzes platform dominance, streaming trends, and the relationship between streams and physical sales
- **Global vs Korea Dashboard:** Examines market segmentation and export performance
- **Label Dashboard:** Evaluates label-level performance, market share, and strategic positioning

10.2 Key Visualization Elements

The dashboard incorporates various visualization types to effectively communicate insights:

- **Line Charts:**
Used to display trends over time, such as sales growth
- **Bar Charts:**
Used for comparing performance across artists, labels, and categories
- **Donut Charts:**
Used to represent proportional distributions, such as market share and segmentation
- **Scatter Plots:**
Used to analyze relationships between variables, such as streaming vs sales
- **KPI Cards:**
Used to highlight key metrics for quick interpretation

10.3 Interactive Features

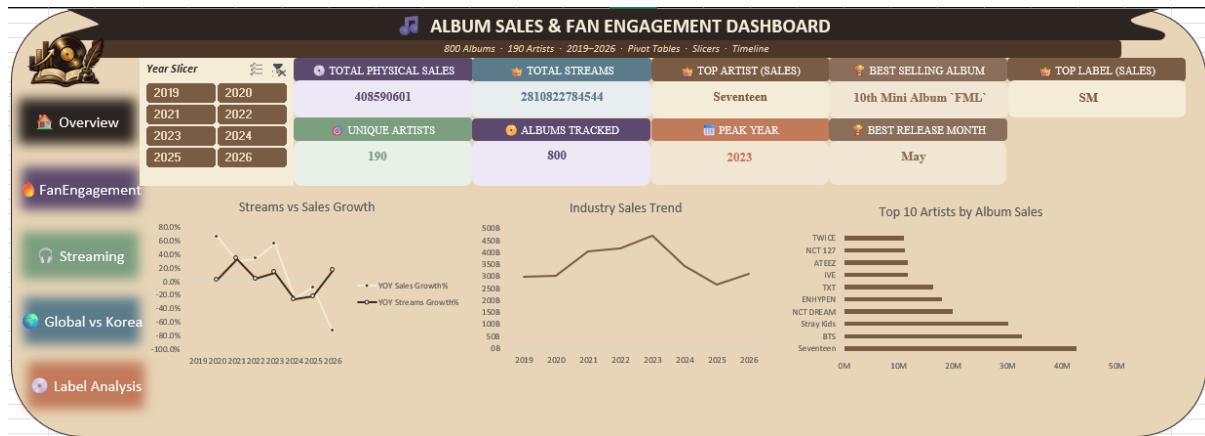
To enhance usability and analytical flexibility, the dashboard includes interactive components:

- **Slicers:** Allow users to filter data dynamically by year, label, and category
- **Dynamic Pivot Tables:** Enable real-time updates across all visualizations

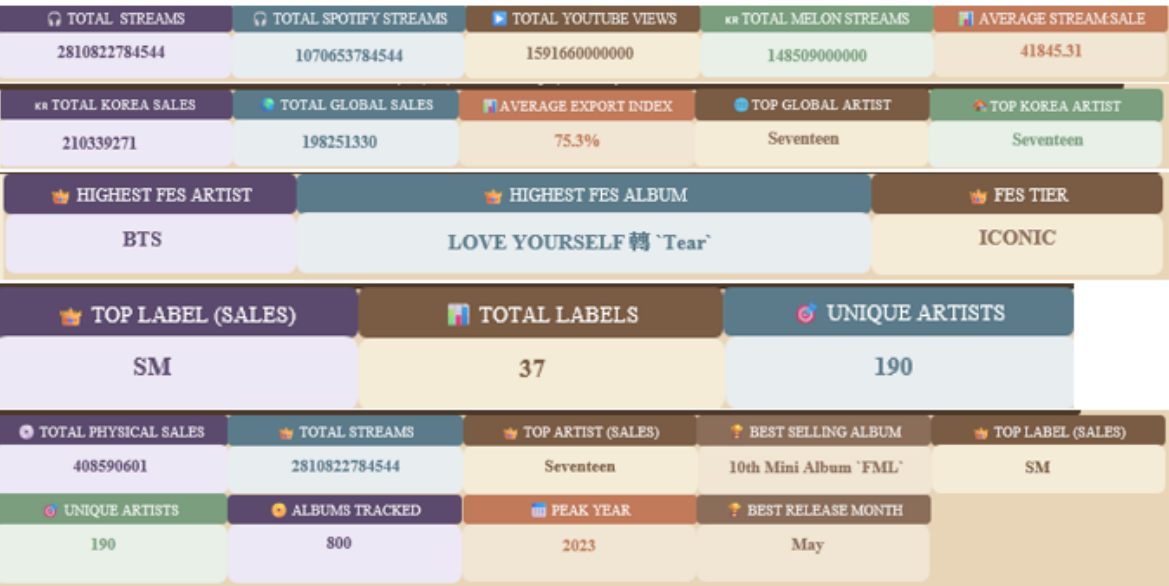
These features allow users to explore data from multiple perspectives without modifying the underlying dataset.

10.4 Automation & Scalability

The architecture is built for sustainability. Because the transformations are handled via **Power Query M-Code**, the entire dashboard can be updated with new monthly sales or streaming data by simply refreshing the connection. This ensures the documentation and the tool itself remain relevant beyond the initial project lifecycle.



DASHBOARD OVERVIEW



KPI OVERVIEW

11. CONCLUSION & FUTURE OUTLOOK

11.1 Conclusion

This project successfully demonstrates how data analytics can be applied to transform fragmented music industry data into a structured and insight-driven analytical system. By integrating album sales, digital streaming, and fan engagement metrics, the project provides a comprehensive understanding of artist performance and market dynamics.

A key contribution of this work is the development of the **Fan Engagement Score (FES)**, which enables unified evaluation across multiple dimensions of performance. The implementation of normalization, weighting logic, and tier classification allows for meaningful comparison and segmentation of artists and albums.

The multi-dashboard framework—comprising Overview, Fan Engagement, Streaming, Global vs Korea, and Label Analysis—provides a layered analytical perspective. This structure allows users to move from high-level summaries to detailed insights, supporting both exploratory analysis and decision-making.

11.2 Key Findings

- **The Global Shift:** With an average Export Index of **75.3%**, the project confirms that international markets have surpassed domestic consumption as the primary driver of artist growth.
- **The Power of Engagement:** The **Fan Engagement Score (FES)** successfully identified that high-tier success is more heavily influenced by consistent digital and social presence (89% of the weighted score) than by physical sales alone.
- **Market Concentration:** A significant performance gap remains between the **12 "Legend" tier artists** and the rest of the industry, suggesting that global dominance is increasingly concentrated among a select few elite performers.
- **Future Growth Velocity:** Projections for **2024–2026** indicate that while the market is stabilizing, the "Rising" artist segment (115 artists) remains the most critical area for portfolio diversification and label investment.

11.3 Future Outlook

While the current project provides a comprehensive analytical framework, several opportunities exist for further enhancement and expansion:

- **Real-Time Data Integration:** Incorporating live data sources and APIs to enable real-time dashboard updates
- **Predictive Analytics:** Applying machine learning models to forecast album performance, fan engagement, and market trends
- **Deployment on Advanced Platforms:** Migrating the dashboard to tools such as Power BI or web-based platforms for improved scalability and accessibility

These enhancements would further strengthen the analytical capabilities of the system and enable more advanced decision-support applications.